BARINDER SINGH BANWAIT

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ABOUT

Research mathematician and software engineer working at the interface of AI and (formalized) mathematics. I build computational tools for number theory research, and am increasingly focused on machine-verifiable proofs and strongly-typed systems for representing mathematical knowledge.

Industry experience in robotics and quantitative finance, alongside machine learning projects on both sensor and mathematical data. Contributor to open-source mathematical software, with current interests in formal verification and AI for mathematics.

SKILLS

Programming Languages

Python | C/C++ | Lean 4 | Sage | Magma | Matlab | PARI/GP

Machine Learning

PyTorch | TensorFlow | scikit-learn | pandas | NumPy

Formal Verification

Lean 4 | Mathlib

Tooling & Infrastructure

Git | Bitbucket | SVN | AWS | Gurobi | Elasticsearch | Kibana | TeamCity

Domains

Number Theory | Arithmetic Geometry | Formal Mathematics | Robotic Control | Quantitative Finance

FORMER ACADEMIC POSITIONS

Lodha Mathematical Sciences Institute
Nov 2025 – Mar 2026 — Mumbai, India

Massachusetts Institute of Technology
Sep 2024 – Aug 2025 — Cambridge, MA, USA

Boston University
Sep 2022 – Aug 2025 — Boston, MA, USA

Ruprecht-Karls-Universität Heidelberg
Oct 2021 – Apr 2022 — Heidelberg, Germany

Harish-Chandra Research Institute
Feb – Sep 2021 — Prayagraj, India

Max-Planck-Institut für Mathematik
Oct 2016 – Nov 2016 — Bonn, Germany

Universität Duisburg-Essen
Jan 2015 – Jan 2017 — Essen, Germany

Institut national de recherche en informatique et en automatique (INRIA)
Jan – Dec 2014 — Bordeaux, France

FORMER INDUSTRY POSITIONS

Quantitative Analyst

Quantile
Sep 2019 – Mar 2020 — London, UK

- Linear, mixed-integer, and multi-objective optimisation for compression of interest-rate derivative portfolios using Gurobi.
- Visualisation of FX trading datasets across several client investment banks.
- Modelling of reset risk and PV01 for swaptions.
- Git code management with Bitbucket.

Research Engineer

CMR Surgical

Jan 2018 – Sep 2019 | Cambridge, UK

- Research and optimisation of robotic control algorithms, including inverse kinematics and mass-spring-damper models.
- Path planning for collision-free end-effector motion.
- Mathematical modelling in Matlab, with Robotics and Control Systems toolboxes.
- Writing production-level, safety-critical embedded C code, compliant with MISRA C and International Standard IEC 62304.
- Time-series telemetry processing of sensor data in Python, using pandas, numpy, and matplotlib.
- Analysis and visualisation of system log messages with Elasticsearch and kibana.
- Development with Amazon Web Services, including Lambda, S3, and Athena.
- Implementing machine learning algorithms for robot arm condition monitoring, using PyTorch and Tensorflow.
- Unit and Regression tests in C, C++, and Matlab, including Google Test framework, continuously integrated with TeamCity.
- Agile software development with SVN and Git.

EDUCATION

PhD Mathematics

University of Warwick

Jan 2010 – Sep 2013 | Coventry, UK

Supervisor: Prof. John Cremona

BA and MMath Mathematics

University of Cambridge - Christ's College

Oct 2005 – June 2009 | Cambridge, UK

MMath (*Part III of the Mathematical Tripos*) - Distinction

PAPERS AND PREPRINTS

1. **A formal proof of the Ramanujan–Nagell theorem in Lean 4**. *arXiv:2604.09808* (2026).
2. **On the Visibility category of the Shafarevich–Tate group**, with J. Caro and S. Chidambaram. Submitted, *arXiv:2601.21519* (2026).
3. **On the Identification of Elliptic Curves That Admit Infinitely Many Twists Satisfying the Birch–Swinnerton-Dyer Conjecture**, with X. Huang. To appear, Algorithmic Number Theory Symposium XVI *(2026)*.
4. **Machine Learning Approaches to the Shafarevich-Tate Group of Elliptic Curves**, with A. Babei, A. Fong, X. Huang and D. Singh. *Advanced in Theoretical and Mathematical Physics*. 29(8) 2353-2379 (2025).
5. **Torsion subgroups of elliptic curves over quadratic fields and a conjecture of Granville**, with M. Derickx. *Algorithmic Number Theory Symposium XVI* (2024).
6. **Towards strong uniformity for isogenies of prime degree**, with M. Derickx. Submitted, *arXiv:2302.08350* (2024).
7. **Computing nonsurjective primes associated to Galois representations of genus 2 curves**, with A. Brumer, H. J. Kim, Z. Klagsbrun, J. Mayle, P. Srinivasan and I. Vogt. *Contemporary Mathematics 796* (2023).
8. **Modularity over $\mathbb{C}$ implies modularity over $\mathbb{Q}$**. To appear, *Contemporary Mathematics* (2023). (Proceedings of *Modularity and the generalized Fermat equation*.)
9. **Explicit isogenies of prime degree over number fields**, with M. Derickx. *Algebra and Number Theory*, 19(6), 1147–1197 (2025).
10. **Cyclic isogenies of elliptic curves over fixed quadratic fields**, with F. Najman and O. Padurariu. *Mathematics of Computation*. Volume 93, 841–862 (2023).
11. **Explicit isogenies of prime degree over quadratic fields**. *International Mathematics Research Notices*. 2023(14):11829–11876 (2023).
12. **Examples of abelian surfaces failing the local-global principle for isogenies**. *Research in Number Theory*. 7(55) (2021)

13. **Correction: Examples of abelian surfaces failing the local-global principle for isogenies**. *Research in Number Theory.* 8(98) (2022)
14. **Del Pezzo surfaces over finite fields and their Frobenius traces**, with F. Fité and D. Loughran. *Mathematical Proceedings of the Cambridge Philosophical Society.* 167(1) (2019) 35-60.
15. **Tetrahedral Elliptic Curves and the local-global principle for isogenies**, with J. Cremona. *Algebra and Number Theory.* 8:5 (2014) 1201-1229.
16. **On some local to global phenomena for abelian varieties**. PhD Thesis, University of Warwick (2013).

RESEARCH PROJECT LEADER

Effective Algebra and the LMFDB

Jan 2025 — Kampala, Uganda

Leading project *L-functions and Galois representations* at CIMPA Schools Uganda.

Rethinking Number Theory 5

June 2024 — Online

Leading project *Machine-Learning the rank and Shafarevich-Tate group of a rational elliptic curve*.

CONFERENCE AND WORKSHOP ORGANISER

- *Arithmetic geometry with a view toward computation*, JMM Special session, San Francisco, CA, Jan 2024
- *Young Researchers in Mathematics*, University of Warwick, Apr 2011

OPEN SOURCE SOFTWARE CONTRIBUTIONS

Formalizing the Ramanujan–Nagell Theorem - Lean 4, Mathlib4

2026

Formally verifying that the equation $x^2 + 7 = 2^n$ has only 10 pairs of solutions (x, n) in the integers.

L-Functions and Modular Forms Database - Python, Postgres, MongoDB

2020–2025

Data management and UI across Elliptic Curves, Classical and Bianchi Modular Forms, Testing utilities, Dirichlet Characters, and Modular Curves.

Absolutely simple endomorphism rings - SageMath, Python

2021

First functionality to check for geometric simplicity of Jacobians of genus 2 curves over $\mathbb{Q}$. Released in sage-9.5.

TEACHING EXPERIENCE

Course Lecturer

Computational Number Theory

Oct 2021 - Feb 2022 — Heidelberg, Germany

Masters course covering algorithmic and computational topics in elliptic curves, modular forms, and algebraic number theory.

Vertiefung Zahlentheorie

Apr - July 2016 — Essen, Germany

Representability of primes via quadratic forms - from Fermat, Euler, Gauss, and to Artin Reciprocity. Three hours per week for 15 weeks. Lectures given in German.

Einführung in das Computer-Algebra-Paket *Sage*

Sep 2015 — Essen, Germany

Introductory week-long course on *Sage* aimed at final year undergraduates. Course given in German.

Algebraic Number Theory

July 2009 | Linyi, China

Introductory course at summer school aimed at second year undergraduates.

Seminar Organiser

Prime numbers and Cryptography

Apr 2022 | Heidelberg, Germany

Bachelor's level seminar organised with Sriram Chinthalagiri

Abelian Varieties

Oct 2021 - Feb 2022 | Heidelberg, Germany

Masters level seminar organised with Prof. Böckle.

Algebraic Surfaces

Oct 2015 - Jan 2016 | Essen, Germany

Masters level seminar organised with Prof. Görtz.

Teaching Assistant

Linear Algebra	Apr - July 2015	Vytautas Paškūnas	Universität Duisburg-Essen
Modular Forms	Oct - Dec 2012	Mehmet Haluk Şengün	University of Warwick
Algebraic Number Theory	Jan - Mar 2012	Johan Bosman	University of Warwick
Elliptic Curves	Oct - Dec 2011	Lassina Dembélé	University of Warwick
Algebraic Number Theory	Jan - Mar 2010	William Hart	University of Warwick

Undergraduate Supervisor

Oct 2010 - Apr 2013 | University of Warwick

Holding supervisions of groups of 5 undergraduates.

DIVERSITY OUTREACH

Raising a Mathematician Workshop

Feb 2026 | Mumbai, India

Working with Mumbai-based *Raising a Mathematician* foundation to promote passion for mathematical education in India.

CIMPA Schools

Jan 2025 | Kampala, Uganda

Project funded by French government to promote mathematical research in developing countries.

Rethinking Number Theory 5

June 2024 | Online

An AIM Research community to stimulate research by people from under-represented groups and to discuss changing norms of the profession.

Cambridge–Linyi Summer school

June 2009 | Linyi, China

Teaching Algebraic Number Theory to students at Linyi Normal University in Shandong, China.

STEP mentor

Apr 2007 – 2009 Cambridge, UK

Coaching groups of A-Level students in the *Sixth Term Examination Papers* mathematics entrance exams to increase diversity and access at Cambridge.

REFEREE DUTIES FOR JOURNALS

Mathematics of Computation | International Journal of Number Theory | Algebra and Number Theory | Research in Number Theory | Acta Arithmetica | International Mathematics Research Notices | Algorithmic Number Theory Symposium | Expositiones Mathematicae

AWARDS

Engineering and Physical Sciences Research Council, UK

Jan 2010

- Full funding for PhD studies.

Institute for Advanced Study, Princeton NJ, USA

Jul 2008

- Full funding to attend Undergraduate Summer School Program of Park City Mathematics Institute on Algebraic Geometry in 2008.

Whelan Prize

Oct 2007

- Awarded by Christ's College, University of Cambridge, for outstanding examination performance (top of college in mathematics).

Nuffield Trust, UK

Jul 2007

- Undergraduate Research Bursary to conduct summer research project.
- Supervisor: Dr. Jon Bevan, University of Surrey, UK.

LANGUAGES

English | ਪੰਜਾਬੀ (Punjabi) | Deutsch | हिंदी (Hindi) | Français

REFEREES

Prof. John Cremona *University of Warwick*
@ J.E.Cremona@warwick.ac.uk

Prof. Andrew Sutherland *Massachusetts Institute of Technology*
@ drew@math.mit.edu

Prof. Jennifer Balakrishnan *Boston University*
@ jbala@bu.edu

RIGHT TO WORK

United Kingdom | India